# Calibration of Production Transient Alert Brokers: Preparing Decision Infrastructure for LSST

*Can broker confidence scores be trusted for automated follow-up decisions?*

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QR code

Code, tables & figures
branch **alerce-selective-deferral**

## 1 Motivation

**"When a broker reports 80% confidence, is it actually correct 80% of the time?"**

Millions of alerts → Broker probabilities → Gates & rankings → Limited follow-up

- Rubin/LSST alert volumes cannot be inspected manually.
- A classifier can rank the correct class first while still reporting an unreliable confidence.

**Underconfidence**
Correct predictions score too low → useful objects fail fixed gates.

**Overconfidence**
Wrong predictions score too high → unreliable labels auto-accepted.

**A functioning classifier is not automatically functioning decision infrastructure.**

## 2 Data & ground truth

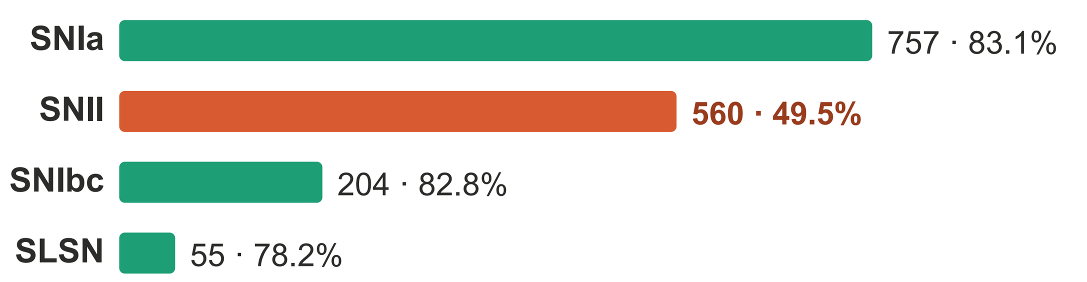
Spectroscopically confirmed ZTF Bright Transient Survey objects, cross-matched against the production outputs of three brokers. BTS spectroscopy is the independent ground truth.

**1,576**
ALeRCE in-taxonomy transients (15-class)

**3**
production broker systems audited

| Broker score | Task | N |
|---|---|---|
| ALeRCE light-curve RF | 15-class | 1,576 |
| NEEDLE | 3-class | 429 |
| Fink SuperNNova / RF | binary SNIa | 1,368 |

### ALeRCE class counts and accuracy




bar length = sample size · label = top-1 accuracy

Overall ALeRCE accuracy 70.9%, but strongly class-dependent — **SNII is the classification bottleneck**.

## 3 Method

1. **Score correctness.** Top-ranked class vs independent spectroscopic label.
2. **Audit calibration.** Does reported confidence match observed accuracy?

$$\mathrm{ECE} = \Sigma_b\,(n_b/N)\cdot|\,\mathrm{acc}(b) - \mathrm{conf}(b)\,|$$

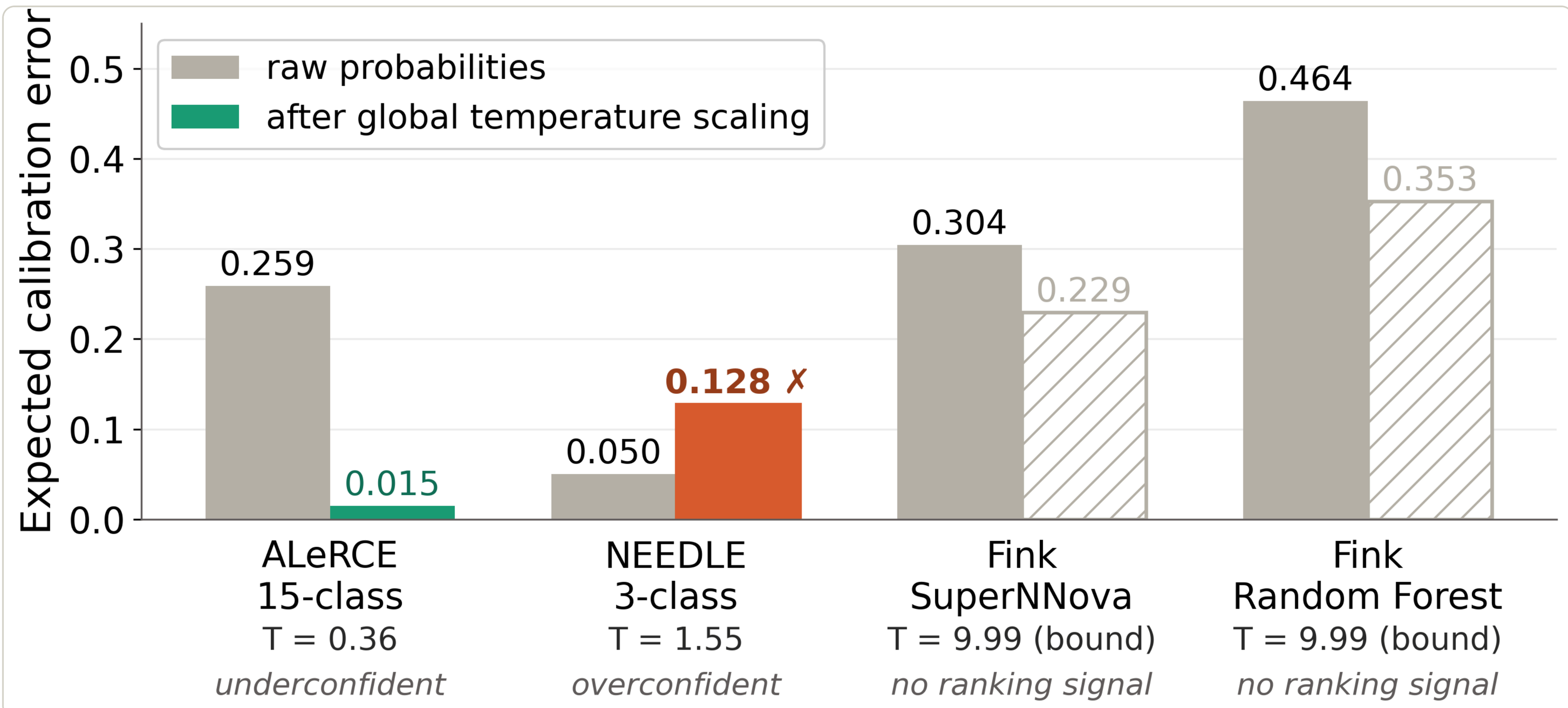
3. **Recalibrate.** Held-out temperature scaling. Brokers expose probabilities, so log-probabilities act as pseudo-logits and softmax returns a valid simplex.

$$\tilde{p} = \mathrm{softmax}(\,\log p\,/\,T\,)\quad \text{monotonic — top-1 class unchanged}$$

4. **Train a reliability checker.** Logistic model estimating **P(top-1 correct)** from the calibrated vector plus its shape (margin, entropy, branch mass), benchmarked against a predicted-class-rate control.
5. **Accept or defer.** Rank by estimated reliability; defer the least trustworthy fraction.

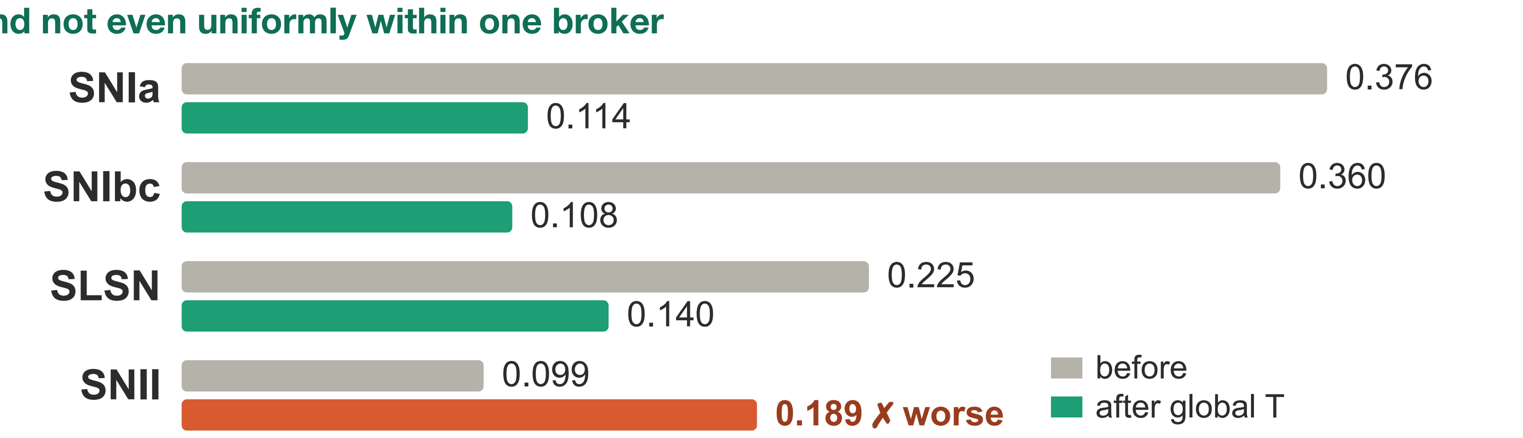
Five-fold out-of-fold evaluation · paired stratified bootstrap · ROC-AUC, AURC and risk–coverage. *Deferral means more photometry, a second broker or human review — not automatically spectroscopic.*

## 4 Three brokers, three failure modes




The same global correction behaves differently for every system. **ALeRCE** is strongly underconfident (T < 1) and is repaired. **NEEDLE** starts best in aggregate but is *overconfident*, and the identical procedure **makes it worse**. For **Fink**, T runs to the optimiser bound — hatched bars mark a nominal drop that is not a usable gain.

### ...and not even uniformly within one broker




Per-class ALeRCE ECE. SNIa and SNIbc are repaired; **SNII degrades** — the aggregate improves while one class gets worse, motivating class-conditional calibration.

## 5 Results — repairing and using the ALeRCE scale

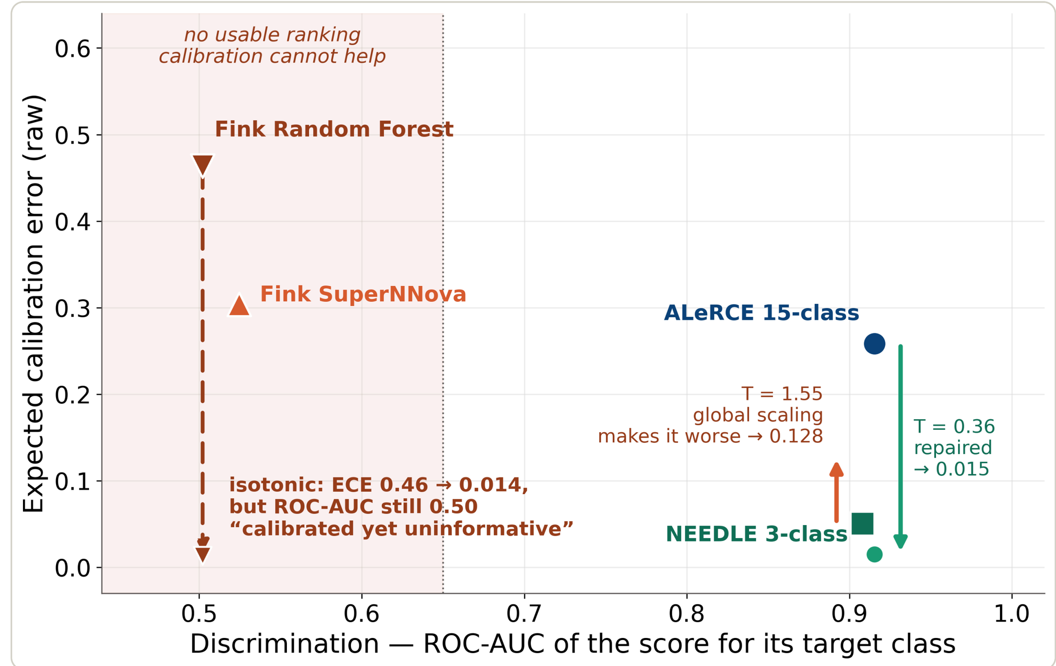
**AGGREGATE CALIBRATION**
**0.259 → 0.015**
ECE, held-out temperature scaling (T ≈ 0.36).

**FIXED GATE P ≥ 0.8**
**9 → 395**
Objects accepted at the same threshold, 88.4% purity — a 44-fold change.

**CHECKER RANKING**
**0.836**
ROC-AUC. Class-rate control 0.752 · raw confidence 0.689.

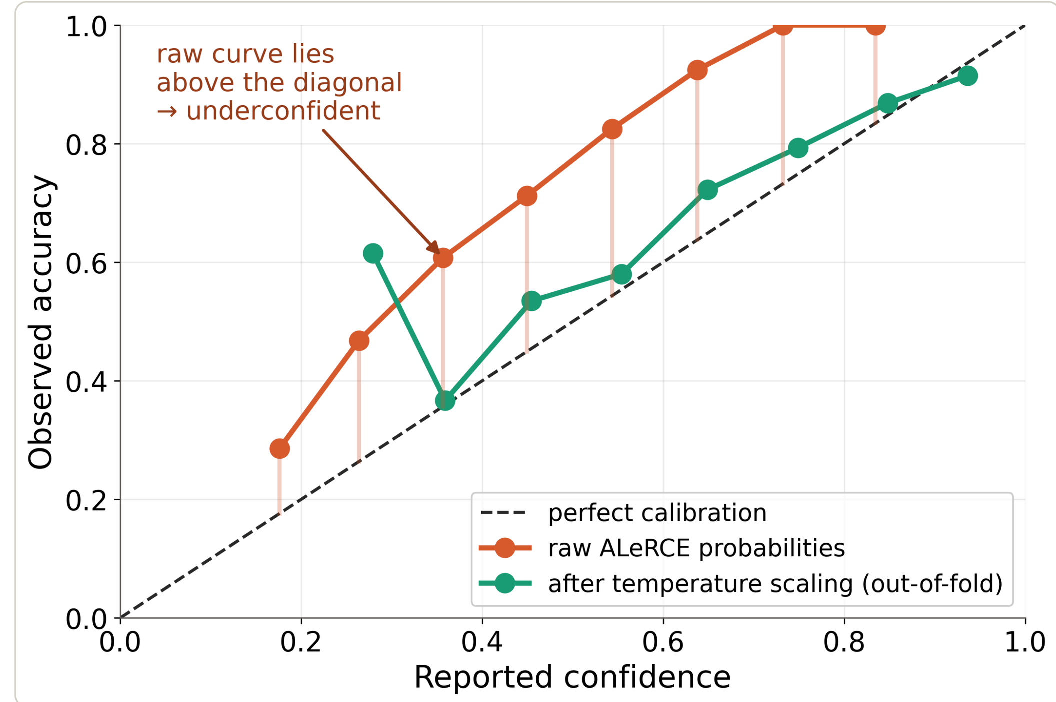
**AT 80% AUTO-ACCEPT**
**81.0% · 47.8%**
Accepted accuracy · errors captured in the deferred 20%.

### Calibration only helps a score that already ranks



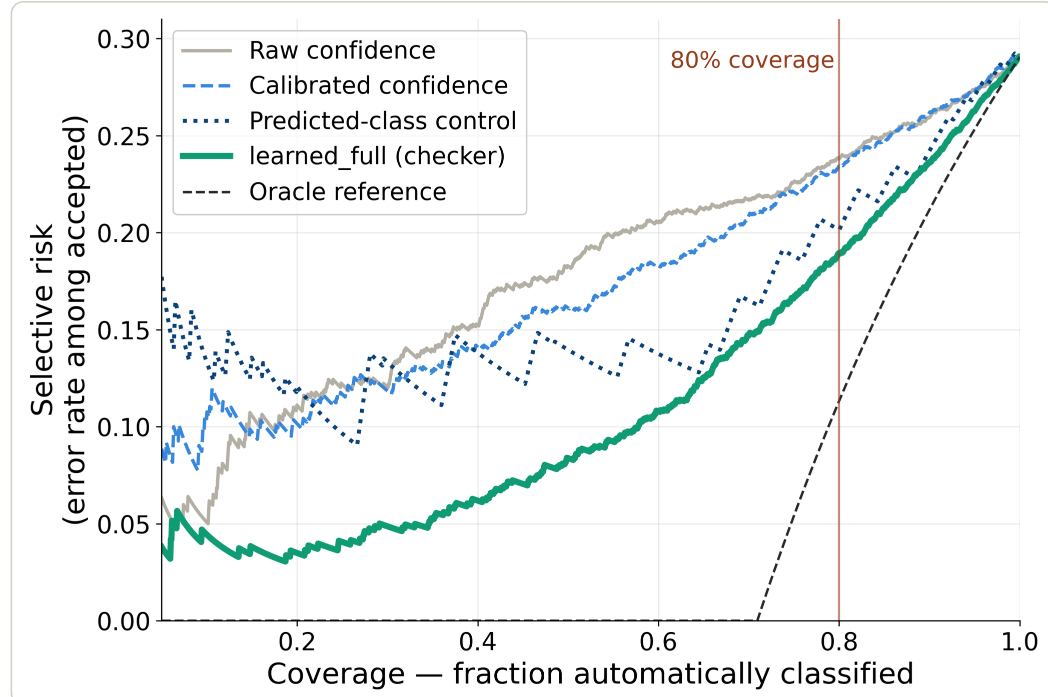

Fink RF reaches **ECE 0.014 under isotonic regression while its ROC-AUC stays at 0.50** — perfectly calibrated and completely uninformative. **A low ECE is not a certificate of usefulness.**

### ALeRCE is systematically underconfident



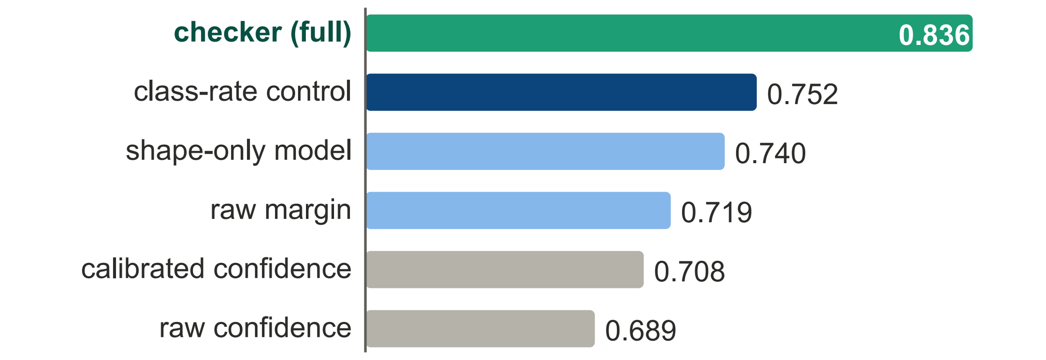

Raw ALeRCE sits far above the diagonal — correct far more often than it claims. Temperature scaling pulls it onto the diagonal *without changing any predicted label.*

### Selective classification




| Policy @ 80% coverage | Acc. | Errors caught |
|---|---|---|
| Raw confidence | 76.2% | 34.5% |
| Predicted-class control | 79.9% | 44.5% |
| **Checker (full vector)** | **81.0%** | **47.8%** |

### The checker beats class identity alone




Δ = **+0.083** over the class-rate control, paired-bootstrap 95% CI [0.057, 0.107] — object-level reliability beyond class identity. The shape-only model does *not* clear the control: the signal is in the full 15-class vector.

## PROPOSED Calibration-aware spectroscopic deferral

Broker probabilities
↓
Calibration layer
↓
Checker risk $r_i$ = P(classification is wrong)
↓
Scientific value & observability
↓
**Top-B targets for spectroscopy**

$$\max_z \Sigma_i\, z_i\, r_i\, V_i\, q_i \quad \text{s.t.} \quad \Sigma_i\, z_i\, c_i \le B$$

$z_i$ select for spectroscopy · $r_i$ checker error risk · $V_i$ scientific value · $q_i$ probability of a usable spectrum · $c_i$ exposure cost · $B$ observing budget. In words: **choose the candidates where a spectrum buys the most science within the available budget.** A separate novelty channel will flag objects outside the broker taxonomy.

Not implemented or tested here — these results *motivate* a future calibration-aware learning-to-defer framework.

## 6 Conclusions

**1.** Production brokers fail in different directions: ALeRCE is strongly underconfident, NEEDLE overconfident, and the audited Fink scores carry almost no ranking signal.

**2.** Calibration repairs what a probability *means* — it never changes the predicted class, but it decides which objects clear a fixed gate.

**3.** A low ECE is not a certificate: a score can be made near-perfectly calibrated while remaining uninformative.

**4.** A checker reading the full probability vector identifies unreliable predictions beyond predicted-class identity, concentrating errors into a small deferred queue.

**Calibration** fixes the scale. A **reliability checker** decides which predictions to trust. Neither is optional if broker outputs are to drive automated follow-up.

**A functioning alert classifier is not automatically functioning decision infrastructure.**

## Main references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